Varun Shirbhayye

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Education

Indian Institute of Technology (IIT) Bhilai Jul 2025 – Jul 2027
M.Tech in Data Science and Artificial Intelligence *CGPA: 7.88 / 10.00*

Indian Institute of Information Technology (IIIT) Nagpur Jul 2018 – Jul 2022
B.Tech in Computer Science and Engineering *CGPA: 8.33 / 10.00*

Industry Experience

IBM India Pvt. Ltd. Jan 2022 – Jun 2022
Project Trainee - Intern *Bengaluru, India*

- Developed Python-based automation scripts to streamline operational data workflows, reducing repetitive manual processing tasks and improving execution consistency across the analytics team.
- Built data preprocessing and validation pipelines using **Pandas** and **NumPy** to support large-scale business process automation; worked with IBM DB2 for schema management and query optimisation.
- Designed Tableau dashboards tracking key business KPIs for cross-functional stakeholders, translating raw data into actionable analytical reports.

Projects & Publications

IIT Bhilai CoPilot | *Python, FastAPI, LangGraph, Docker, Qdrant, SQLite, Agentic AI*

- Built a multilingual RAG assistant for verified information retrieval from official public IIT Bhilai sources
- Combined dense retrieval, BM25, Reciprocal Rank Fusion, and cross-encoder reranking, achieving **92.75% Hit@5** and **0.8533 MRR**.
- Implemented LangGraph-based tool routing, structured fact freshness and conflict resolution, citation verification, prompt-injection protection, and privacy-preserving conversation memory.

PGARC: Persona-Gated Adversarial Representation Learning | *Python, PyTorch, Transformers, NLP*

- Developed an adversarial representation-learning framework to reduce persona recoverability while preserving semantic content in dialogue representations.
- Designed a learned persona subspace with adaptive gating and evaluated it across DistilBERT, GPT-2, and BART.
- Benchmarked representations using five post-hoc attackers and controlled baselines including GRL, HSIC, INLP, LEACE, and projection-based methods across multiple random seeds.

Vehicle Fuel Efficiency Prediction | *Python, Scikit-learn, Pandas* Paper link

- [**IEEE Published**] Co-authored "Accurate Prediction of MPG using Linear Regression in Machine Learning," published in *IEEE Xplore Conference Proceedings*, 2020.
- Built a linear regression model on the UCI Auto-MPG dataset to predict fuel efficiency, achieving an R^2 score of 0.82.
- Performed feature engineering, cross-validation, and residual analysis to evaluate model performance and identify key factors affecting MPG.

Technical Skills

Languages: Python, C, C++, SQL, NoSQL, HTML, CSS, Javascript, SQLite

Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, OpenCV, FastAPI, Docker

Platforms: Git, Google Colab, VS Code, Windows, Linux

Soft Skills: Leadership, Insight Synthesis, Creative Problem Solving, Agile Prioritization, Good Communication

Achievements

- **Top 4 globally** — HackerEarth Machine Learning Challenge: Exhibit A(rt) (competitive ML, global leaderboard).
- **Qualified** — Google Code Jam 2021 Qualifier Round (global competitive programming contest).
- **Rank 1382** — Google Kick Start 2019, Round F .
- **Rank 1141 / 8051** — HackerRank HackFest 2020.

Research interests

- Machine Learning, Deep learning, Computer Vision, and Natural Language Processing and Development.